

Multi-User Physiological Computing

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CS 495
Preliminary Presentation

Sponsor & Team Members



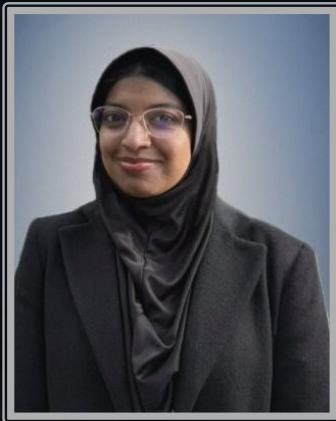
Dr. Chris Crawford



Trenton Ottman



Ansel Stinnett



Zara Ahmed



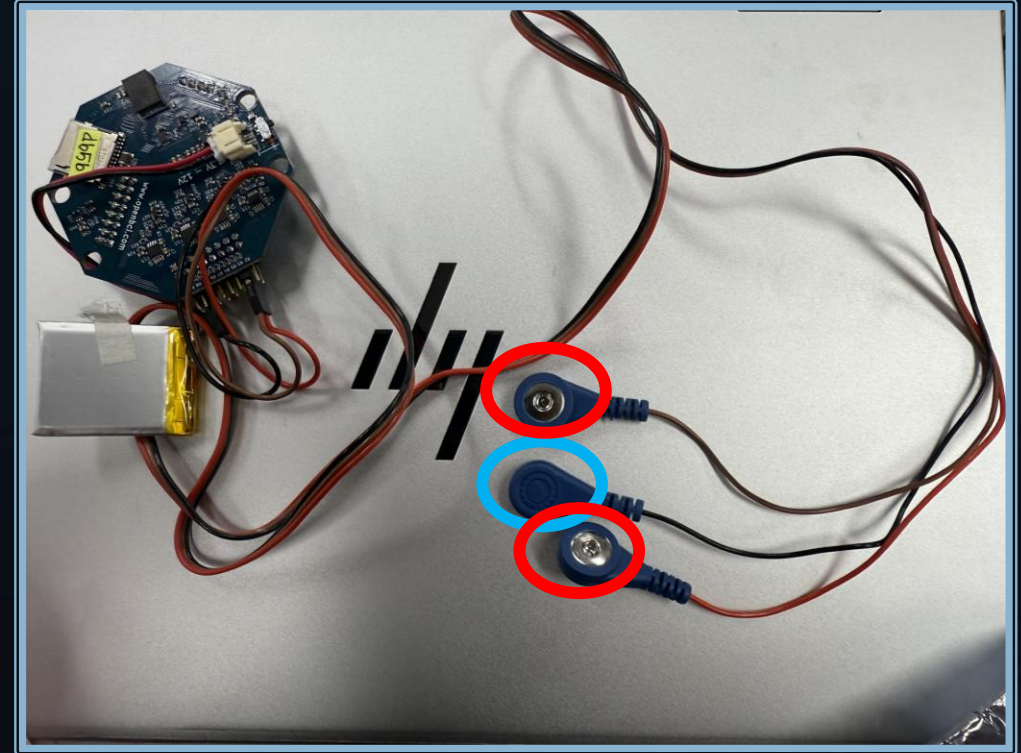
Jackson Wallace



James Adler

Introduction

- Physical sensors to be used in browser-based gaming
- The problem/task is to implement real-time data streaming from multiple sensors and design a game for 8-12 players
- End-users are K-12 students



Proposed Solution

- Solution: Develop a web-based multiplayer game using physiological computing
- Initial Scope
- Major Project Backlog Themes (multiplayer and real-time, game design, game functionality)
- Highest Priority Issues

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OVERHEAT

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Current Project Overview

- Done: Read input from physiological device.
- Done: Singleplayer game integration.
- TODO: Decide what to keep from old project.
- TODO: Investigate potential tech stacks.
- TODO: Prototype real-time multipoint server.

Strategy & Workflow

- Quality/strategy
 - Pull request review before every merge
 - Server tested with 12 simulated players
 - Play testing as real acceptance test
- Workflow
 - Two week sprints, with a meeting with sponsor every cycle
 - Done = merge, reviewed and demo able
- Risks / Challenges
 - May not have 8-12 devices → build a fake sensor feed to develop against
 - Server may lag with many players → develop under load early

Dependencies & Unknowns

➤ Dependencies

- Currently exploring 3rd party dependencies such as Playroom
- Completed code from sponsor
- Depending on chosen stack more dependencies are possible

➤ Unknowns

- What type of game will be made
- Current cybersecurity risks



Questions?